

Use of Antifibrinolytics in Pediatric Life-Threatening Hemorrhage: A Prospective Observational Multicenter Study

OBJECTIVES: To assess the impact of antifibrinolytics in children with life-threatening hemorrhage.

DESIGN: Secondary analysis of the MAssive Transfusion epidemiology and outcomes In Children study dataset, a prospective observational study of children with life-threatening bleeding events.

SETTING: Twenty-four children's hospitals in the United States, Canada, and Italy.

PATIENTS: Children 0–17 years old who received greater than 40 mL/kg of total blood products over 6 hours or were transfused under activation of massive transfusion protocol.

INTERVENTION/EXPOSURE: Children were compared according to receipt of antifibrinolytic medication (tranexamic acid or aminocaproic acid) during the bleeding event.

MEASURES AND MAIN RESULTS: Patient characteristics, medications administered, and clinical outcomes were analyzed using Cox proportional hazard and Kaplan-Meier survival analysis. The primary outcome was 24-hour mortality. Of 449 patients analyzed, median age was 7 years (2–15 yr), and 55% were male. The etiology of bleeding was 46% traumatic, 34% operative, and 20% medical. Twelve percent received antifibrinolytic medication during the bleeding event ($n = 54$ unique subjects; $n = 18$ epsilon aminocaproic acid, $n = 35$ tranexamic acid, and $n = 1$ both). The antifibrinolytic group was comparable with the non-antifibrinolytic group on baseline demographic and physiologic parameters; the antifibrinolytic group had longer massive transfusion protocol duration, received greater volume blood products, and received factor VII more frequently. In the antifibrinolytic group, there was significantly less 6-hour mortality overall (6% vs 17%; $p = 0.04$) and less 6-hour mortality due to hemorrhage (4% vs 14%; $p = 0.04$). After adjusting for age, bleeding etiology, Pediatric Risk of Mortality score, and plasma deficit, the antifibrinolytic group had decreased mortality at 6- and 24-hour postbleed (adjusted odds ratio, 0.29 [95% CI, 0.09–0.93]; $p = 0.04$ and adjusted odds ratio, 0.45 [95% CI, 0.21–0.98]; $p = 0.04$, respectively).

CONCLUSIONS: Administration of antifibrinolytic medications during the life-threatening event was independently associated with improved 6- and 24-hour survivals in bleeding children. Consideration should be given to use of antifibrinolytics in pediatric patients with life-threatening hemorrhage.

KEY WORDS: aminocaproic acid; antifibrinolytic agents; blood transfusion; child; hemorrhage; tranexamic acid

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Bleeding is a major cause of preventable death in pediatric trauma, surgical, and medical patients (1–3). Mortality rates in children with life-threatening bleeding have been reported to range from 24% to 65% depending on the underlying etiology of the bleed (4–6), higher than reported mortality rates for nonbleeding pediatric cohorts with comparable illness severity (8–16%) (7–9).

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Hemostatic resuscitation has evolved over recent decades to include high-ratio blood product transfusion, prevention of coagulopathy, hypothermia and acidosis, and surgical or interventional control of bleeding when applicable (10, 11). The majority of resuscitation practice guidelines in pediatrics are derived from adult data despite the fact that children's coagulation and physiologic responses to bleeding unequivocally vary from that of adults (12, 13). This is a result of a lack of high-quality data in children with life-threatening bleeding. Pediatric-specific data are desperately needed to guide therapies to improve outcomes in children with life-threatening bleeding.

Intrinsic abnormalities of fibrinolysis are well-documented and are associated with poor outcomes in bleeding children (14–16). There are increasing data, particularly in the trauma and obstetric literature, to support reduced mortality in those patients who receive antifibrinolytics in their acute resuscitation (17, 18). As a result, antifibrinolytics are standard of care for bleeding trauma and obstetric patients, and have been integrated into many adult massive transfusion protocols (MTPs) (19, 20). Small, single-site studies in pediatric patients with traumatic, medical, or surgical bleedings suggest that there may be benefit to either tranexamic acid (TXA) or epsilon aminocaproic acid (EACA) for bleeding children (21–24). However, there are no studies of comparable quality and scale to the adult literature to guide the use of antifibrinolytics in contemporary guidelines for pediatric life-threatening hemorrhage. This secondary analysis of a large multicenter prospective observational study of children with life-threatening bleeding assesses the impact of antifibrinolytics administered during the life-threatening bleeding on early survival.

MATERIALS AND METHODS

Study Design and Data

This secondary analysis of the MAssive Transfusion epidemiology and outcomes In Children (MATIC) study database was planned a priori. A full description of the MATIC study has been published elsewhere (4). Briefly, this was a prospective, observational study of children with life-threatening bleeding at 24 medical centers in the United States, Canada, and Italy from January 2014 to October 2018. Inclusion criteria were children 0–17 years old with life-threatening hemorrhage. Discrete bleeding events were identified and recorded by the

contributing center. These were defined by MTP activation with blood product transfusion, or if child received greater than 40 mL/kg of any blood product (25), or combination of blood products for active bleeding, in a 6-hour period with no break in blood product administration greater than 30 minutes. A subject could have multiple bleeding events by these criteria if sufficient time elapsed between blood product transfusions or if a new MTP was activated. For patients with multiple bleeding events, only the first event was analyzed.

Data extracted included demographics, etiology of bleeding (trauma, operative, and medical), Glasgow Coma Scale (GCS), vital signs, measures of end-organ function, fluids, blood products, hemostatic adjuncts, Pediatric Risk of Mortality score (PRISM III) (26), injury severity score, morbidities, and mortality. The PRISM III score was calculated from data available within 24 hours prior to the life-threatening bleeding and includes: systolic blood pressure, diastolic blood pressure, heart rate, respiratory rate, $\text{PaO}_2/\text{FIO}_2$, PaCO_2 , GCS score, pupillary reaction, prothrombin time, partial thromboplastin time, total bilirubin, potassium, calcium, glucose, and bicarbonate. When calculating PRISM III scores, per the validated methods, all missing values were assumed to be normal (26). Missing weight values were estimated based on age and sex from the Centers for Disease Control and Prevention Clinical Growth Charts. Cause of death was determined by the participating sites (27). The study was conducted under waiver of informed consent granted by each participating center's institutional review board (**Supplemental Digital Content**, <http://links.lww.com/CCM/G889>).

Analysis

Data were summarized as counts, median (interquartile range [IQR]), or percentage. Subjects were categorized into the antifibrinolytic group if they received either TXA or EACA during their life-threatening bleeding event. Unadjusted analyses included Wilcoxon rank-sum and Kruskal-Wallis testing for skewed continuous data and chi-square or Fisher exact test for categorical data. The main outcomes were 6- and 24-hour survivals; 28-day survival was assessed as a secondary outcome (28, 29). Kaplan-Meier analysis was used to calculate the cumulative probability of 6-hour, 24-hour, and 28-day survival as a function of time. Log-rank testing for equality of survivor functions was then performed. Cox proportional-hazards

regression modeling was performed to determine independent predictors of 6-hour, 24-hour, and 28-day mortalities using all variables with $p < 0.2$ on univariate analysis on univariate analysis and those determined a priori to be clinically significant. Variables were chosen for the final model using forward stepwise selection to minimize Bayesian information criteria. All hypothesis tests were two-sided, and a p value of less than 0.05 was considered significant. All statistical analyses were conducted using the STATA software, Version 16 (StataCorp, College Station, TX).

RESULTS

Among 449 subjects analyzed, 55% were male, and the median age was 7 years (2–15 yr). The etiology of bleeding was 34% traumatic, 46% operative, and 20% medical. The mortality rates for the whole cohort were as follows: 6-hour mortality = 15% ($n = 69$), 24-hour mortality = 22% ($n = 99$), and 28-day mortality = 38% ($n = 168$). Subjects were categorized based on receipt of antifibrinolytic medication. In the antifibrinolytic group, 12% received EACA and/or TXA as part of their hemostatic resuscitation ($n = 54$ unique subjects; $n = 18$ aminocaproic acid, $n = 35$ TXA, and $n = 1$

received both). Antifibrinolytics were administered in the majority of participating sites (65% [23/35]) without statistically significant variation in the number of antifibrinolytic recipients between the sites ($p = 1.00$).

The duration of life-threatening bleeding for patients who did not receive an antifibrinolytic was 196 minutes (75–338 min). The duration of life-threatening bleeding for patients who received an antifibrinolytic was 383 minutes (189–551 min). Time from initiation of the life-threatening bleeding to TXA administration was median (IQR) = 66 minutes (22–177 min), and a median (IQR) total dose of 20 mg/kg (14–43 mg/kg) of TXA was administered. The dosing regimens included: single bolus ($n = 17$), bolus followed by infusion ($n = 7$), and infusion only ($n = 12$). Time from the initiation of the life-threatening bleeding to EACA administration was median (IQR) = 106 minutes (30–199 min), and a median (IQR) total dose of 100 mg/kg (54–148 mg/kg) of EACA was administered. Seven children received a second dose of EACA in the same bleeding event.

The antifibrinolytic group was comparable with the no-antifibrinolytic group on baseline characteristics including age, sex, race, ethnicity, body mass index, and etiology of the bleed, as well as vital signs and baseline laboratory values prior to the life-threatening bleed

TABLE 1.
Comparison of Antifibrinolytic and No-Antifibrinolytic Groups

Measures	Antifibrinolytic ($n = 54$)	No Antifibrinolytic ($n = 395$)	p
Demographics			
Age (yr) ($n = 449$)	8 (3–16)	7 (1.5–15)	0.12
Sex ($n = 449$)			
Male	26 (48%)	221 (56%)	0.31
Female	28 (52%)	174 (44%)	
Race ($n = 380$)			
White	31 (61%)	211 (64%)	0.76
Black	15 (29%)	94 (29%)	
Other	5 (10%)	24 (7%)	
Ethnicity: Hispanic/Latino ($n = 392$)	7 (15%)	36 (10%)	0.33
Body mass index ($n = 321$)	19 (16–23)	19 (15–23)	0.90
Blood group ($n = 436$)			
A	17 (31%)	140 (37%)	0.79
B	9 (17%)	56 (15%)	
AB	2 (4%)	10 (3%)	
O	26 (48%)	176 (46%)	

(Continued)

TABLE 1. (Continued).
Comparison of Antifibrinolytic and No-Antifibrinolytic Groups

Measures	Antifibrinolytic (n = 54)	No Antifibrinolytic (n = 395)	p
Pre-life-threatening hemorrhage variables			
Pediatric Risk of Mortality score ^a (n = 428)	18 (7–22)	12 (6–22)	0.48
Indication for transfusion (n = 449)			
Operative	230 (37%)	133 (34%)	0.66
Trauma	22 (41%)	185 (47%)	
Medical	12 (22%)	77 (19%)	
Lowest Glasgow Coma Scale score (n = 280)	7 (3–15)	8 (3–15)	0.79
Lowest temperature (n = 339)	36 (35.2–36.5)	36.2 (35.2–36.6)	0.38
Age-adjusted hypotension (n = 443)	22 (41%)	166 (43%)	0.88
Age-adjusted tachycardia (n = 422)	32 (64%)	231 (62%)	0.88
Hematocrit (n = 420)	29 (25–35)	29 (24–35)	0.93
International normalized ratio (n = 384)	1.5 (1.2–1.8)	1.4 (1.2–1.8)	0.44
Platelet count (n = 395)	146 (93–240)	163 (99–250)	0.32
Base deficit (n = 375)	–1 (–6 to 3)	0.6 (–5 to 6.0)	0.16
Blood products and adjuncts			
Duration of massive transfusion protocol (n = 449)	383 (189–551)	196 (75–338)	< 0.001
Total volume, mL/kg (n = 449)	89 (42–216)	56 (28–112)	< 0.001
RBC volume, mL/kg (n = 449)	39 (18–83)	30 (15–59)	0.07
Plasma volume, mL/kg (n = 449)	28 (13–73)	16 (4–33)	< 0.001
Platelet volume, mL/kg (n = 449)	14 (7–29)	7 (0–17)	< 0.001
Cryoprecipitate volume, mL/kg (n = 449)	0 (0–5)	0 (0–1.6)	0.007
Plasma deficit (n = 449)	10 (–2–35)	12 (2–30)	0.71
Platelet deficit (n = 449)	23 (8–60)	22 (9–45)	0.39
Plasma:RBC ratio (n = 413)	0.62 (0.48–0.98)	0.50 (0.17–0.84)	0.001
Platelet:RBC ratio (n = 413)	0.33 (0.16–0.54)	0.19 (0–0.38)	< 0.001
Crystalloid volume mL/kg	23 (3–49)	18 (0–52)	0.37
Factor VII (yes/no) (n = 447)	16 (29%)	40 (10%)	< 0.001

^aThe Pediatric Risk of Mortality III score includes systolic blood pressure, diastolic blood pressure, heart rate, respiratory rate, $\text{Pao}_2/\text{Fio}_2$, Paco_2 , Glasgow Coma Scale score, pupillary reaction, prothrombin time, partial thromboplastin time, total bilirubin, potassium, calcium, glucose, and bicarbonate.

Numeric values are median (interquartile range) when continuous and n (%) when discrete.

(Table 1). Regarding the bleeding event, those in the antifibrinolytic group had a longer duration of MTP activation, were transfused a greater weight-adjusted volume of RBCs, plasma and platelets, and more often received factor VII as a hemostatic adjunct. There was no difference between the antifibrinolytic and no-antifibrinolytic groups in the location of the bleeding event ($p = 0.96$), which was most commonly in the emergency department (35% vs 37%, respectively),

operating room (27% vs 31%, respectively) or ICU (35% vs 31%, respectively). Regarding outcome (Table 2), the antifibrinolytic group had significantly lower 6-hour mortality than the nonantifibrinolytic group (6% vs 17%; $p = 0.04$) and lower 6-hour mortality due to hemorrhage (4% vs 14%; $p = 0.04$) on bivariate analysis. There was no statistical difference with the use of antifibrinolytics compared with the nonantifibrinolytic group for 24-hour mortality (13% vs 23%;

TABLE 2.
Outcomes

Measures	Antifibrinolytic (n = 54)	No Antifibrinolytic (n = 395)	p
Mortality–6 hr (n = 69/449)	3 (6%)	66 (17%)	0.04
Death due to bleeding (n = 54)	2 (4%)	52 (14%)	0.04
Death due to CNS injury (n = 13)	1 (2%)	12 (2%)	> 0.99
Death due to other (n = 2)	0 (0%)	2 (1%)	> 0.99
Mortality–24 hr (n = 99/449)	7 (13%)	92 (23%)	0.11
Death due to bleeding (n = 71)	5 (10%)	66 (18%)	0.17
Death due to CNS injury (n = 23)	2 (4%)	21 (6%)	0.75
Mortality–28 d (n = 173/449)	20 (38%)	148 (38%)	1.00
Acute respiratory distress syndrome (n = 449)	13 (24%)	78 (20%)	0.47
Sepsis (n = 449)	6 (11%)	38 (10%)	0.80
Acute kidney injury (n = 449)	16 (29%)	67 (17%)	0.02

$p = 0.11$) or 28-day mortality (38% vs 38%; $p > 0.99$). Survival on Kaplan-Meier time to event analysis was significantly improved in 6-hour mortality ($p = 0.04$)

and approached significance in 24-hour mortality ($p = 0.08$) (**Fig. 1**); there was no difference in 28-day mortality ($p = 0.67$). The antifibrinolytic group had a higher

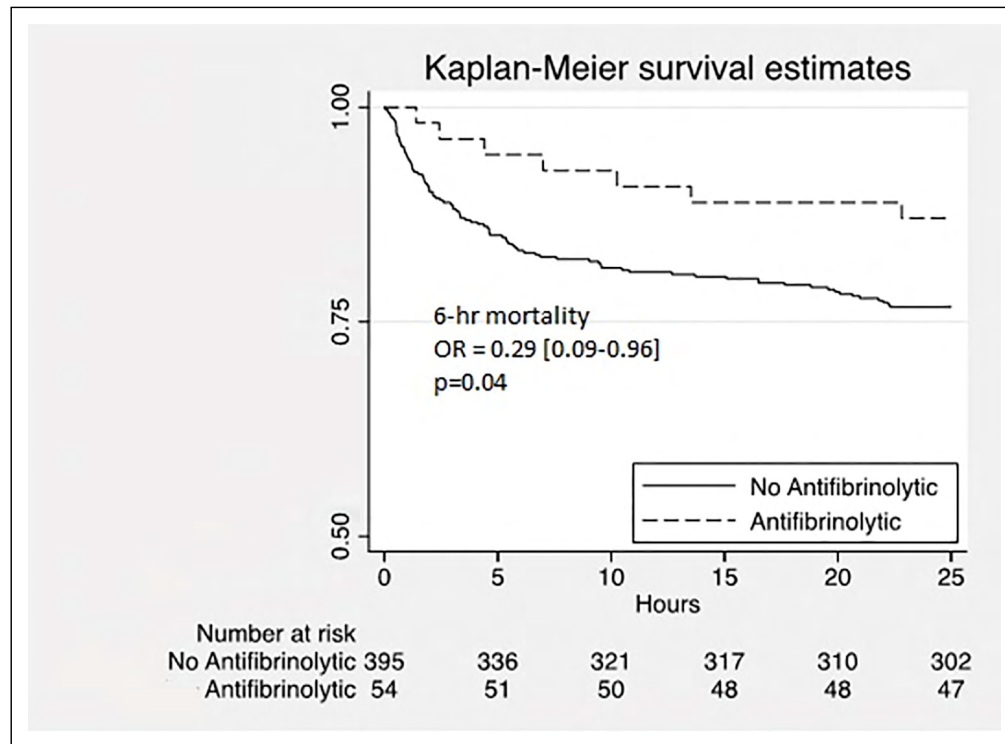


Figure 1. Kaplan-Meier survival curve: resuscitation with antifibrinolytics is associated with improved 6-hr ($p = 0.04$) and approached significance for improved 24-hr mortality ($p = 0.08$) on unadjusted analysis compared with resuscitation without antifibrinolytics. Both 6-hour ($p = 0.04$) and 24-hour ($p = 0.04$) survivals were improved after adjusting for Pediatric Risk of Mortality score, etiology of the bleed, age, and plasma deficit. OR = odds ratio.

occurrence rate of acute kidney injury (AKI; 29% vs 17%; $p = 0.02$); this difference was attributable to the EACA group (10/19, 52%) compared with the TXA group (7/36, 19%). There was no difference between the groups in a composite outcome of death or AKI (49% vs 56%; $p = 0.3$).

Regarding the adjusted analyses, in the Cox proportional-hazards regression model accounting for age, etiology of the bleeding, and PRISM score, children who received antifibrinolytic medications as part of their resuscitation had significantly reduced adjusted odds ratio (AOR) of mortality at 6- and 24-hour postbleed (**Table 3**). The covariates that were

TABLE 3.
Cox Proportional Hazard Models for 6- and 24-Hour Mortalities

Measures	6-hr Mortality		24-hr Mortality	
	OR (95% CI)	<i>p</i>	OR (95% CI)	<i>p</i>
Unadjusted analysis				
Antifibrinolytic	0.29 (0.09–0.96)	0.04	0.49 (0.21–1.12)	0.08
Adjusted analysis				
Antifibrinolytic	0.29 (0.09–0.93)	0.04	0.45 (0.21–0.98)	0.04
Pediatric Risk of Mortality score	1.04 (1.02–1.06)	< 0.001	1.05 (1.04–1.07)	< 0.001
Indication operative (ref)				
Trauma	4.18 (1.90–9.17)	0.001	3.08 (1.71–5.56)	< 0.001
Medical	5.45 (2.42–12.26)	< 0.001	3.28 (1.91–6.60)	< 0.001
Age (yr)	0.94 (0.90–0.98)	0.002	0.95 (0.92–0.98)	0.002
Plasma deficit	1.01 (0.99–1.01)	0.17	1.01 (0.99–1.01)	0.12

OR = odds ratio.

Other variables considered in model generation: sex, age-adjusted hypotension, base deficit, total blood product volume, RBC volume, plasma volume, platelet volume, cryoprecipitate volume, platelet deficit, plasma:RBC ratio, platelet:RBC ratio, crystalloid volume, recombinant Factor VIIa.

considered and ultimately excluded in the process of generating a model with optimal fit were participating center, sex, age-adjusted hypotension, base deficit, total blood product volume, RBC volume, plasma volume, platelet volume, cryoprecipitate volume, platelet deficit, plasma:RBC ratio, platelet:RBC ratio, crystalloid volume, and use of recombinant Factor VIIa. Although antifibrinolytic administration did not differ significantly by site on bivariate analysis, site was included in each model as a sensitivity analysis to evaluate for center effect. Enrolling center was not significantly associated with mortality at any time point, and model fit was superior without center as a covariate; therefore, center was excluded from the final models. The 28-day survival was not significantly different between the antifibrinolytic versus no-antifibrinolytic groups (AOR = 0.78; 95% CI, 0.49–1.26; *p* = 0.32) nor was rate of second life-threatening hemorrhage event (6% vs 10%, respectively; *p* = 0.2). Unadjusted odds ratios for subgroups (bleeding indication and choice of antifibrinolytic medication) for both 6-hour (Fig. 2) and 24-hour mortalities were in general of similar magnitude and direction compared with the entire cohort but did not reach statistical significance. The one subgroup that did not follow the trend of unadjusted odds ratios were the patients with operative bleeding at 24-hour mortality.

DISCUSSION

In this secondary analysis of a multicenter prospective pediatric study, antifibrinolytics were independently associated with decreased at 6- and 24-hour mortalities when used as a hemostatic adjunct early, within a median of 1–2 hours, from the initiation of life-threatening hemorrhage. Notably, death due to bleeding 6-hour posthemorrhage was reduced. This is the first multicenter prospective analysis to support the effectiveness of antifibrinolytics in reducing early death if administered in the acutely bleeding pediatric patient.

The mechanisms by which antifibrinolytics contribute to hemostasis and improve survival are complex and remain incompletely characterized, especially in children. TXA and EACA are both synthetic lysine analogs that competitively bind the lysine-binding sites on plasminogen, thus inhibiting the conversion of plasminogen to plasmin by tissue plasminogen activator and urokinase and limiting binding to the fibrin surface. Both agents may also directly inhibit the activity of plasmin at certain doses (30). The choice of agent is often dictated by geography, hospital formulary, and clinician preference. These medications differ by side-effect profile: TXA has been associated with seizures in a dose-dependent manner (31–33), and EACA has been associated with postprocedural renal dysfunction

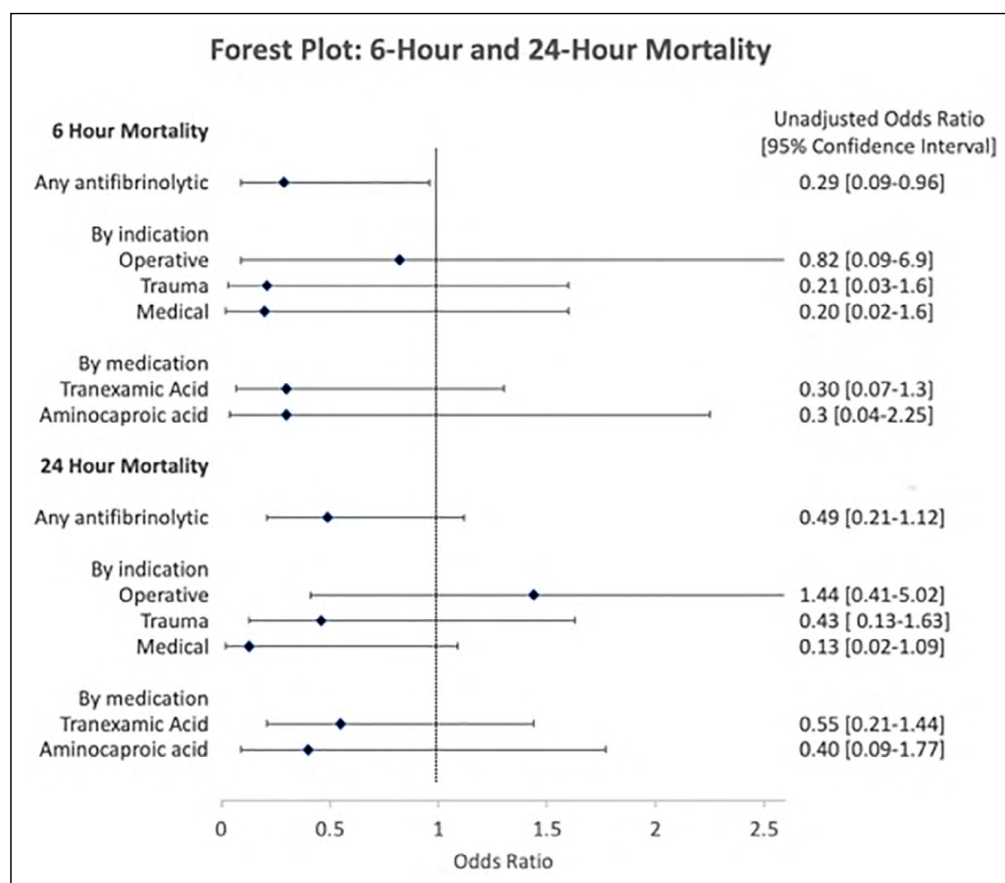


Figure 2. Forest plot of 6- and 24-hr mortality unadjusted odds ratios for overall cohort and by subgroups: bleeding etiology (trauma, operative, and medical) and antifibrinolytic medication choice (tranexamic acid or epsilon aminocaproic acid).

(34), as reported in an unadjusted analysis in this study. There are no definitive data demonstrating superiority of either agent, and both have been shown to reduce bleeding and improve patient outcomes (23, 34, 35).

The data presented here are the first from a multicenter prospective study to report a survival benefit independently associated with antifibrinolytics in a cohort of children with various bleeding etiologies. Overall, there is a lack of data regarding the use of antifibrinolytics in pediatric cohorts with hemorrhage and, therefore, little to guide providers in the care of these patients. In a recent systematic review, there were too few quality studies to render definitive guidance (21). Perhaps due to the lack of guidelines or a national standard, the use of antifibrinolytics in pediatric centers remains relatively uncommon compared with that in adult centers (36, 37). Retrospective cohort studies have suggested benefits of both EACA and TXA (38–40) but are limited by sample size and study design. There are several single-center randomized controlled studies that report on the effectiveness of

antifibrinolytics in select pediatric populations. In a placebo-controlled trial of 43 children undergoing craniosynostosis reconstruction surgery, the TXA group had significantly lower perioperative mean blood loss (65 vs 119 mL/kg; $p < 0.001$) and lower perioperative mean blood transfusion (33 vs 56 mL/kg; $p = 0.006$) compared with the placebo group (22). A randomized controlled trial of 150 children undergoing cardiac corrective surgery on cardiopulmonary bypass compared TXA, EACA, and placebo; the placebo group had the longest sternal closure time, greater blood loss at 24 hours, and greater requirements of blood and blood products. There was no difference between TXA and EACA

groups in postoperative blood loss, blood and product requirement, and reexploration rates (23). In a military cohort of injured pediatric patients in Afghanistan, the 66 children who received TXA within 3 hours of injury had decreased mortality, reduced need for ventilatory support, and improved neurologic status without increased thromboembolic complication compared with injured children who did not receive TXA (24). In a subsequent study including a select military cohort of injured pediatric patients requiring massive transfusion (> 40 mL/kg in 24 hr), a mortality benefit was seen for TXA recipients ($n = 59$) compared with those who did not receive TXA ($n = 448$) after adjusting for age, sex, head injury severity, and base deficit (OR, 0.35; 0.12–0.99; $p = 0.048$) (41).

The data in the present study are concordant with the antifibrinolytic benefits seen in these pediatric studies. In our study, at baseline, the antifibrinolytic group was comparable with the no-antifibrinolytic group on demographics, prebleeding vital signs, and laboratory values. The antifibrinolytic group did receive increased

volume of RBCs, plasma, and platelets; these factors were eliminated in the process of generating a parsimonious model with optimal fit, as their relative importance to the outcome was low. This difference in blood product administration is likely attributable to survival bias. Children who lived longer during the life-threatening bleeding and had greater duration of MTP received greater weight-based volume of blood product resuscitation. AKI as an outcome is predisposed to similar issues with survival bias, as survivors will appear to develop increased late complications; to account for survivor bias, a composite outcome of mortality and AKI indicated no differences between the study groups.

Recently, an international panel of experts in trauma care recommended that early mortality outcomes are clinically relevant and should be considered for primary outcomes of pivotal trials. In this National Institutes of Health and Department of Defense-sponsored workshop, 6- and 24-hour mortalities were recommended as primary outcomes for studies of hemostatic agents in children with severe traumatic bleeding, as early mortality is considered a reflection of the efficacy of an intervention in severely bleeding patients (28, 29). Adoption of these early mortality outcomes has been accepted by the Food and Drug Administration as primary outcomes in pivotal trials in both children and adults (personal communication, P.C.S.). In comparison, late mortality (at 28 d) is considered to be a safety-related outcome instead of an efficacy-related outcome in trials of hemostatic agents in severely bleeding patients since the effect of the product will occur within 24 hours of administration (29).

In the present study, the similar 28-day mortality rate between the study groups is not unexpected. Late death could reasonably be attributable to progression of pre-existing conditions (e.g., malignancy) or the underlying cause of the bleed, factors that would not be addressed by antifibrinolytic administration during the hemostatic resuscitation. This dataset does not include cause of death with sufficient granularity to determine which were potentially “preventable” deaths. Nevertheless, children with life-threatening bleeding have a very high mortality rate (4, 42); improving early survival after an acute bleeding episode is valuable, as this provides an opportunity to use other therapies that may mitigate or target factors associated with late mortality.

The etiology of bleeding is an important factor in adult studies of antifibrinolytics. Although the

recommendation for early administration of antifibrinolytics is supported by robust data in adult trauma (17, 18) and obstetric (32, 43) patients, the benefit of antifibrinolytics in adults with gastrointestinal bleeding is less clear. Despite several meta-analyses that have reported a survival benefit, decreased endoscopic intervention, and decreased bleeding (44, 45), the recently published Haemorrhage ALleviation with Tranexamic acid - Intestinal system (HALT-IT) trial found no difference in death due to GI bleeding (46). Compared with trauma and obstetric cohorts, the HALT-IT cohort was substantially older; importantly, the effect of age on the hemostatic response to antifibrinolytics has yet to be investigated. Our study was limited by an insufficient number of subjects for a robust, multivariable analysis of subgroups by etiology of the bleed. A pilot study comparing TXA with placebo in injured children was completed with results pending (47) and may provide additional data and guidance in this subgroup. Increased survival in general was seen in the current study across indications (medical, surgical, and trauma) and antifibrinolytic choice (EACA and TXA), though these subgroups were too small to permit a robust statistical analysis. Multicenter clinical trials in children with life-threatening bleeding are necessary to confirm the benefit of antifibrinolytics for the various etiologies of hemorrhage.

Limitations of this study include those inherent to secondary analysis of existing data, though notably this particular research question was specified a priori. Very few institutions in this dataset used functional coagulation assays (thromboelastography and thromboelastometry) during the resuscitation; as the availability and use of these assays increase, further study should be done to understand the impact of antifibrinolytic medications in children with various fibrinolysis phenotypes. A related limitation is the lack of standard interval, serial resuscitation labs that would reflect coagulation and physiologic derangement in response to the antifibrinolytic medication; however, this limitation is the one inherent to all studies of hemorrhagic shock where patients require rapid interventions, and it would be inappropriate to delay treatment in order to obtain blood work. The diverse nature of bleeding etiology included in the analysis increases generalizability of results in children with life-threatening hemorrhage. However, the small size of each subgroup did not permit a robust analysis of individual etiologies of

bleeding, and further studies are required to confirm these findings in each subgroup that are suggested by the concordant direction and magnitude of effect size seen in Figure 2. Similarly, the use of multiple antifibrinolytic agents in this contemporary cohort of bleeding children reflects current real-world practice; however, further studies are required to assess the impact of individual medications (TXA vs EACA). Although seizure was not reported in this dataset, this medication side effect is infrequently associated with TXA administration. Occurrence rates for seizure (0–2.7%)(31–33, 48) are dose-dependent, with high-dose regimens conferring greater risk of seizure than low-dose regimens (49, 50). In the present cohort, children received a median TXA dose of 20 mg/kg, consistent with a low-dose regimen; other pediatric cohorts in the literature receiving TXA at a median dose of up to 100 mg/kg, five times this amount, report a 0–0.6% occurrence rate of seizure (48, 51–53). Finally, missing data may predispose to bias; however, the degree of missing data on primary outcome measure was nil and on key variables in the model was minimal (number analyzed reported in Table 1 for each variable) and presumed to be completely at random. Sensitivity analyses where numeric variables were assumed to be in normal range if missing did not differ from complete case analysis, and therefore, complete case analysis is reported.

CONCLUSIONS

Children with life-threatening hemorrhage are at substantial risk for death. Early administration of antifibrinolytic medications as a component of hemostatic resuscitation in bleeding children was independently associated with improved survival at 6- and 24-hour postbleed. Further multicenter randomized controlled trials should confirm these findings, and consideration should be given to use of antifibrinolytics in pediatric patients with life-threatening hemorrhage.

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